Available online at www.sciencedirect.com
ScienceDirect
Current Opinion in Structural Biology

Multiple protein sequence alignment

Jimin Pei

Multiple sequence alignments are essential in computational analysis of protein sequences and structures, with applications in structure modeling, functional site prediction, phylogenetic analysis and sequence database searching. Constructing accurate multiple alignments for divergent protein sequences remains a difficult computational task, and alignment speed becomes an issue for large sequence datasets. Here, I review methodologies and recent advances in the multiple protein sequence alignment field, with emphasis on the use of additional sequence and structural information to improve alignment quality.

Address
Howard Hughes Medical Institute, University of Texas Southwestern Medical Center at Dallas, 5323 Harry Hines Boulevard, Dallas, TX 75390, USA

Corresponding author: Pei, Jimin (jpei@chop.swmed.edu)

Current Opinion in Structural Biology 2008, **18**:382–386

This review comes from a themed issue on
Sequence and Topology
Edited by Nick Grishin and Sarah Teichmann

Available online 14th May 2008

0959-440X/$ – see front matter
Published by Elsevier Ltd.

DOI 10.1016/j.sbi.2008.03.007

Introduction

A variety of sequence and structural analysis methods rely on multiple sequence alignments, including methods for similarity searches, structure modeling, function prediction, and phylogenetic analysis. Construction of a multiple sequence alignment aims at arranging residues with inferred common evolutionary origin or structural/functional equivalence in the same column position for a set of sequences. Position-specific information about residue usage, conservation and correlation can be deduced from a multiple sequence alignment for various applications. Thus, the quality of alignments is a crucial factor for their proper usage. Accurate and fast construction of multiple sequence alignments has been under extensive research in recent years, and a variety of methods have been developed [1–3]. Recent advances in multiple alignment methods trend toward automatic incorporation of various sources of additional information to improve alignment quality. For assessment of alignment quality, methods relying on comparisons to reference alignments and methods using direct structural comparisons on the basis of test alignments are both commonly used and yield consistent results.

Computational methods for multiple protein sequence alignments

While pairwise alignment has simple and tractable algorithms using dynamic programming [4,5], direct extension of these algorithms to aligning multiple sequences is computationally expensive and infeasible for more than a few sequences [6,7]. Therefore, many approximate algorithms have been developed for multiple sequence alignments, including the commonly used progressive alignment technique [8]. Progressive methods assemble a multiple alignment by making a series of pairwise alignments of sequences or pre-aligned groups. The order of these pairwise alignments is guided by a tree or dendrogram so that similar sequences tend to be aligned before divergent sequences. Progressive methods cannot guarantee an optimal solution, and do not correct for errors made in each pairwise alignment step. Using scoring functions based on general residue substitution models, classic progressive methods such as ClustalW [9] are fast and can produce reasonable results for relatively similar sequences (e.g., sequence identity above 30%). However, these methods display limited success in aligning divergent sequences accurately.

To correct or minimize errors made in progressive alignment steps, two techniques are frequently used: iterative refinement and consistency scoring. Iterative refinement is often carried out after progressive assembly of a multiple sequence alignment. This strategy usually involves repeatedly dividing the aligned sequences into sub-alignments and realigning the sub-alignments. With scoring based on general amino acid substation models, MAFFT [10] and MUSCLE [11] are two recent programs that mainly rely on iterative refinement to enhance alignment quality. Fine-tuning of various parameters in progressive methods with iterative refinement is important to achieve optimal results [12].

Exploration of consistency information in progressive alignment was pioneered by the program T-Coffee [13]. The consistency-based scoring function for two sequences depends not only on their pairwise alignments, but also on how they can be aligned with regard to other sequences. For example, for three sequences A, B and C, if residue A_i is aligned to residue B_j and residue B_j is aligned to residue C_k, then this implies that residues A_i and C_k can be aligned through intermediate sequence B. Such transitivity results are taken into account in consistency measures so that the alignment scoring function for two sequences contains information of their alignments to other sequences. For progressive alignments without refinement steps, consistency scoring is superior to scoring based on general amino acid substitution models. Quite a few methods have been developed based on the consistency scheme recently. For example, PCMA [14] measures consistency of sequence profiles instead of individual sequences. ProbCons [15] gives a probabilistic treatment of consistency through pairwise hidden Markov models (HMMs) [16]. The advantage of using HMMs is that they can give an estimation of the probability of any two residues being aligned. Such a technique has yielded additional improvement to consistency-based progressive alignments. In MUMMALS [17•], more complex HMMs are designed that implicitly capture information of unalignable regions, secondary structure and solvent accessibility in probabilistic consistency scoring. ProbAlign [18] and CONTRAlign [19] use different statistical techniques other than HMMs to deduce consistency scoring. One major advantage of the consistency scheme is that it can effectively incorporate different sources of constraints such as local alignments, global alignments and structure-based alignments when available. Consistency-based approaches have also been applied in aligning multiple protein structures [20–22].

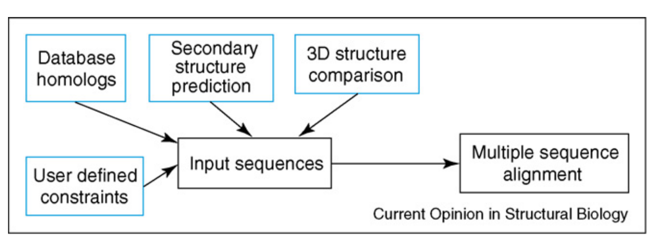
Most available alignment methods assume all the sequences are globally alignable, and they do not perform well for sequences with repeats or different domain architectures. Low-complexity or disordered regions can also cause alignment problems, since the concept of alignable positions does not apply for them. POA [23] and ABA [24] handles the cases of repeats or shuffled domains better by representing alignments using more informative graphic models. ProDA [25•] is another program that is specifically designed to deal with repeats and shuffled domains by exhaustive searching of locally alignable regions among sequences. Global trace graph [26••] is an approach that organizes non-redundant representatives of all known protein sequences into a graph of aligned positions based on consistency and transitivity of locally alignable residues, which has been effective in searching for distant homologs.

Using additional information improves alignment quality

The current best alignment methods that only explore information in input sequences performs similarly on average, and they all give mediocre alignment quality when sequence similarity falls below the 'twilight zone' (e.g. identity less than 20%). Sequence and structural databases are expanding rapidly owing to genome sequencing projects and structural genomics initiatives, offering helpful sources to further improve multiple protein sequence alignments. Three types of additional information are exploited in some multiple alignment methods: sequence homologs, predicted secondary structures, and known 3-dimensional (3D) structures. Sequence homologs provide more evolutionary information, and allow better estimation of position-specific residue usages (profiles). As structures are generally more conserved than sequences, structural information is also valuable for aligning sequences. Significant boosts to alignment quality, especially for distantly related sequences, have been observed when additional homologs and/or structures are incorporated in alignment process. DbClustal [27] combines local alignments found in database searching with global ClustalW alignments. MAFFT package [28] has a program (Mafft-homologs) that aligns target sequences together with found database homologs to obtain more accurate alignments for the targets. In PRALINE [29], PROMALS [30••] and SPEM [31], database homologs are identified by PSI-BLAST [32] searches to build sequence profiles and predict secondary structures, and profile-to-profile comparisons enhanced with secondary structural information are used in alignment processes. PROMALS implements a probabilistic consistency scoring based on profile–profile comparison HMMs. Using database homologs and predicted secondary structures has resulted in about a 10% increase as compared to purely sequence based methods on the PREFAB [11] benchmark, and even larger accuracy increase on the most difficult 'twilight zone' set of the SABmark benchmark [33].

Under the consistency framework, available 3D structural information can be combined with sequence information in building multiple sequence alignments. 3DCoffee [34] uses SAP [35] structure-based alignments and FUGUE [36] sequence-to-structure alignments to improve alignment quality. Recently, the Expresso server [37] extends the 3DCoffee method by automatically identifying highly similar 3D structural templates for target sequences and using structural alignments for consistency-based alignments. Using a similar approach, PROMALS3D [38] combines structural constraints derived from several state-of-the-art structural comparison programs, with sequence constraints derived from profile–profile comparison with predicted secondary structures (Figure 1). The MAFFT server and PROMALS3D server also allow input of user-defined alignment constraints.

Figure 1



The sources of additional information used in PROMALS3D server.

Although using additional sources of information can increase alignment quality on an average basis, cases exist where errors in these sources can lead to poor alignments. For example, errors in PSI-BLAST alignments of found homologs can result in bad estimations of sequence profiles, secondary structure predictions can be incorrect, and 3D structure comparisons purely from a geometric perspective can contradict with results based on sequence evolutionary models. Thus, using additional sequence and structural information is most beneficial to aligning divergent sequences.

Evaluation of alignment quality

A classic way of alignment quality evaluation is to compare test alignments to reference alignments that are considered to be gold standard (usually structure-based alignments). Commonly used reference alignment benchmarks include BaliBASE [39], HOMSTRAD [40], PREFAB [11], OXBENCH [41] and SABmark [33]. To improve reference quality, alignments in these benchmarks are often manually curated (in the case of BaliBASE and HOMSTRAD) or based on the consensus of different structure comparison programs (PREFAB and SABmark). PREFAB database is noted for containing a large number of alignments (1682 cases in version 4.0). SABmark database is designed for difficult alignment cases with divergent sequences. Although structure-based alignments can serve as high-quality references, they have several drawbacks. First, structural alignments could still have errors, especially for proteins with relatively low structural similarity. Second, defining the optimal structural alignment in certain regions is difficult, and might not be possible for structurally divergent proteins. Third, multiple ways of aligning structurally similar parts exist for multi-domain proteins, but structure comparison programs usually report only one alignment. BaliBASE database has specific sets of references for evaluation of proteins with repeats, transmembrane regions or circular permutations.

Reference-independent evaluation of alignment quality does not require reference alignments and thus avoids many problems associated with them. Reference-independent evaluation compares two protein structures directly using the aligned residue pairs in the test alignment, and calculates scores reflecting the structural similarity of these aligned residues. Structural similarity scores can be based on inter-molecular distances or intra-molecular distances. Scores based on inter-molecular distances, such as RMSD, GDT-TS score [42], TM-score [43] and 3D-score [44], require superposition of aligned residues (usually corresponding C-alpha atoms). Scores based on intra-molecular distances, such Dali Z-score [45], iRMSD [46] and the LiveBench contact scores [44], do not require rigid-body superposition and are more tolerant to domain movements in the structures. Structural similarity scores are reasonable measures of alignment quality since an alignment with better quality should have a higher structural similarity for aligned residues. Although reference-independent evaluations have long been used for assessing structural prediction models [47], their routine use in assessment of multiple sequence alignments is only recent. Reference-independent evaluations using various structural similarity scores produce consistent results compared to reference-dependent evaluations using large alignment benchmarks [17•,46].

For comparison of alignment methods, the average alignment quality scores are usually presented and the statistically significance of differences between methods are reported. An alternative but more direct way to assess performance is to compare individual alignments, and report the number of cases when one method is better than another one and vice versa [17•,48•]. Large-scale comparisons reveal that for distantly related proteins, although different methods such as MUMMALS, ProbCons and MAFFT produce quite similar average accuracy scores, the resulting alignments could be very different in many individual cases [17•]. In addition, a statistically worse method could still outperform a better method on some individual cases. Thus it is a good practice to compare the results of several methods for manual refinement of alignment regions with uncertainty. M-COFFEE [48•] is a meta-method that combines results of several multiple alignment methods using a consistency-scoring scheme. On average, it can produce slightly improved results than the best single method used.

Alignment speed and program selection

The sizes of sequence databases and many large protein families are increasing rapidly. For progressive methods, fast tree building methods have been developed to deal with a large number of sequences [11,49]. Iterative refinement of alignments usually involves many steps of aligning two sub-alignments, and identification and fixing of conserved core blocks can greatly increase speed, as implemented in MUSCLE and MAFFT. Time and memory are much more severe issues for methods that use consistency scoring with operations on sequence triplets. One way of reducing computational time and complexity is to use different strategies for different sets of sequences. Highly similar sequences can be aligned in a fast way without compromising speed. On the other hand, more elaborate techniques and additional sources are required to enhance alignment quality for divergent sequences. In PCMA, similar sequences are aligned in a fast way, and the time consuming consistency scoring is only applied to the relatively divergent pre-aligned groups, the number of which can be much smaller than the number of the original set of sequences. PROMALS and PROMALS3D apply the same strategy to reduce the number of sequences in consistency measure. Recent versions of MAFFT [28] have implemented a simpler

Table 1

Representative multiple sequence alignment programs and their web server sites

Programs using only sequence information	
ClusalW	http://www.ebi.ac.uk/Tools/clustalw/
MAFFT	http://align.bmr.kyushu-u.ac.jp/mafft/online/server/
MUSCLE	http://phylogenomics.berkeley.edu/cgi-bin/muscle/input_muscle.py
ProbCons	http://probcons.stanford.edu/
T-Coffee	http://www.tcoffee.org/
Programs using database homologs and predicted secondary structures	
PROMALS	http://prodata.swmed.edu/promals/
SPEM	http://sparks.informatics.iupui.edu/Softwares-Services_files/spem.htm
Programs using known 3D structures	
Expresso	http://www.tcoffee.org/
PROMALS3D	http://prodata.swmed.edu/promals3d/

and faster consistency measure for gap-free segments in pairwise alignments. For methods that use additional homologs and structures, sequence database searching and comparison of 3D structures can occupy a considerable amount of time. To reduce computation time, PROMALS3D only performs sequence database searching for representative sequences, and makes use of pre-computed structural alignment databases.

Many multiple alignment programs and web servers have been developed, providing users a spectrum of choices. Some programs and their web server addresses are listed in Table 1. Generally, if the target set of sequences are relatively similar to each other, programs using only sequence information such as ClustalW, MAFFT and MUSCLE are good choices for producing reasonable alignment, and they also have the advantage of being fast and able to handle a large number of sequences. However, if special attention is paid to distantly related members, programs using additional information such as SPEM, PROMALS, PROMALS3D and Expresso are more suitable options. The intermediate results produced in these alignment processes, such as secondary structure prediction and found homologs, are also valuable for checking alignment quality and further analysis of the target sequences [50].

Conclusions

Constant developments are seen in the field of multiple sequence alignments, with many methods and web servers appearing in recent years. Iterative refinement and consistency scoring remain the major techniques for improving progressive alignments. Addition of database sequence and structural information has proven effective in enhancing alignment accuracy, especially for aligning distantly related sequences. Servers such as Expresso and PROMALS3D have automated these processes and facilitated the generation of high quality alignments.

Acknowledgements

The author would like to thank Lisa Kinch and James Wrabl for critical reading of the manuscript and helpful comments.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest
- •• of outstanding interest

1. Edgar RC, Batzoglou S: **Multiple sequence alignment**. *Curr Opin Struct Biol* 2006, **16**:368-373.
2. Notredame C: **Recent evolutions of multiple sequence alignment algorithms**. *PLoS Comput Biol* 2007, **3**:e123.
3. Wallace IM, Blackshields G, Higgins DG: **Multiple sequence alignments**. *Curr Opin Struct Biol* 2005, **15**:261-266.
4. Needleman SB, Wunsch CD: **A general method applicable to the search for similarities in the amino acid sequence of two proteins**. *J Mol Biol* 1970, **48**:443-453.
5. Smith TF, Waterman MS: **Identification of common molecular subsequences**. *J Mol Biol* 1981, **147**:195-197.
6. Lipman DJ, Altschul SF, Kececioglu JD: **A tool for multiple sequence alignment**. *Proc Natl Acad Sci U S A* 1989, **86**:4412-4415.
7. Wang L, Jiang T: **On the complexity of multiple sequence alignment**. *J Comput Biol* 1994, **1**:337-348.
8. Feng DF, Doolittle RF: **Progressive sequence alignment as a prerequisite to correct phylogenetic trees**. *J Mol Evol* 1987, **25**:351-360.
9. Thompson JD, Higgins DG, Gibson TJ: **CLUSTALW: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice**. *Nucleic Acids Res* 1994, **22**:4673-4680.
10. Katoh K, Misawa K, Kuma K, Miyata T: **MAFFT: a novel method for rapid multiple sequence alignment based on fast Fourier transform**. *Nucleic Acids Res.* 2002, **30**:3059-3066.
11. Edgar RC: **MUSCLE: multiple sequence alignment with high accuracy and high throughput**. *Nucleic Acids Res* 2004, **32**:1792-1797.
12. Wheeler TJ, Kececioglu JD: **Multiple alignment by aligning alignments**. *Bioinformatics* 2007, **23**:i559-i568.
13. Notredame C, Higgins DG, Heringa J: **T-Coffee: a novel method for fast and accurate multiple sequence alignment**. *J Mol Biol* 2000, **302**:205-217.
14. Pei J, Sadreyev R, Grishin NV: **PCMA: fast and accurate multiple sequence alignment based on profile consistency**. *Bioinformatics* 2003, **19**:427-428.
15. Do CB, Mahabhashyam MS, Brudno M, Batzoglou S: **ProbCons: probabilistic consistency-based multiple sequence alignment**. *Genome Res* 2005, **15**:330-340.

16. Eddy SR: **Hidden Markov models**. *Curr Opin Struct Biol* 1996, **6**:361-365.

17. Pei J, Grishin NV: **MUMMALS: multiple sequence alignment improved by using hidden Markov models with local structural information**. *Nucleic Acids Res* 2006, **34**:4364-4374.
• Complex hidden Markov models are designed to improve probabilistic consistency alignment. It also provides comprehensive alignment evaluations by several techniques.

18. Roshan U, Livesay DR: **Probalign: multiple sequence alignment using partition function posterior probabilities**. *Bioinformatics* 2006, **22**:2715-2721.

19. Do CB, Grosss SS, Batzoglou S: **CONTRAlign: discriminative training for protein sequence alignment**. In *Proceedings of the Tenth Annual International Conference on Computational Molecular Biology (RECOMB 2006)*. 2006.

20. Chen Y, Crippen GM: **An iterative refinement algorithm for consistency based multiple structural alignment methods**. *Bioinformatics* 2006, **22**:2087-2093.

21. Ebert J, Brutlag D: **Development and validation of a consistency based multiple structure alignment algorithm**. *Bioinformatics* 2006, **22**:1080-1087.

22. Konagurthu AS, Whisstock JC, Stuckey PJ, Lesk AM: **MUSTANG: a multiple structural alignment algorithm**. *Proteins* 2006, **64**:559-574.

23. Lee C, Grasso C, Sharlow MF: **Multiple sequence alignment using partial order graphs**. *Bioinformatics* 2002, **18**:452-464.

24. Raphael B, Zhi D, Tang H, Pevzner P: **A novel method for multiple alignment of sequences with repeated and shuffled elements**. *Genome Res* 2004, **14**:2336-2346.

25. Phuong TM, Do CB, Edgar RC, Batzoglou S: **Multiple alignment of protein sequences with repeats and rearrangements**. *Nucleic Acids Res* 2006, **34**:5932-5942.
• A method designed for finding and aligning repeats and different domains for multi-domain proteins.

26. Heger A, Mallick S, Wilton C, Holm L: **The global trace graph, a novel paradigm for searching protein sequence databases**. *Bioinformatics* 2007, **23**:2361-2367.
•• A large scale clustering in protein sequence space by using transitivity of local alignments, useful for sequence database searching.

27. Thompson JD, Plewniak F, Thierry J, Poch O: **DbClustal: rapid and reliable global multiple alignments of protein sequences detected by database searches**. *Nucleic Acids Res* 2000, **28**:2919-2926.

28. Katoh K, Kuma K, Toh H, Miyata T: **MAFFT version 5: improvement in accuracy of multiple sequence alignment**. *Nucleic Acids Res* 2005, **33**:511-518.

29. Simossis VA, Heringa J: **Integrating protein secondary structure prediction and multiple sequence alignment**. *Curr Protein Pept Sci* 2004, **5**:249-266.

30. Pei J, Grishin NV: **PROMALS: towards accurate multiple sequence alignments of distantly related proteins**. *Bioinformatics* 2007, **23**:802-808.
•• Automatic identification of protein homologs and prediction of secondary structures to improve alignment quality, especially for distantly related sequences.

31. Zhou H, Zhou Y: **SPEM: improving multiple sequence alignment with sequence profiles and predicted secondary structures**. *Bioinformatics* 2005, **21**:3615-3621.

32. Altschul SF, Madden TL, Schaffer AA, Zhang J, Zhang Z, Miller W, Lipman DJ: **Gapped BLAST and PSI-BLAST: a new generation of protein database search programs**. *Nucleic Acids Res* 1997, **25**:3389-3402.

33. Van Walle I, Lasters I, Wyns L: **SABmark – a benchmark for sequence alignment that covers the entire known fold space**. *Bioinformatics* 2005, **21**:1267-1268.

34. O'Sullivan O, Suhre K, Abergel C, Higgins DG, Notredame C: **3DCoffee: combining protein sequences and structures within multiple sequence alignments**. *J Mol Biol* 2004, **340**:385-395.

35. Taylor WR: **Protein structure comparison using iterated double dynamic programming**. *Protein Sci* 1999, **8**:654-665.

36. Shi J, Blundell TL, Mizuguchi K: **FUGUE: sequence-structure homology recognition using environment-specific substitution tables and structure-dependent gap penalties**. *J Mol Biol* 2001, **310**:243-257.

37. Armougom F, Moretti S, Poirot O, Audic S, Dumas P, Schaeli B, Keduas V, Notredame C: **Expresso: automatic incorporation of structural information in multiple sequence alignments using 3D-Coffee**. *Nucleic Acids Res* 2006, **34**:W604-W608.

38. Pei J, Kim BH, Grishin NV: **PROMALS3D: a tool for multiple protein sequence and structure alignments**. *Nucleic Acids Res* 2008.

39. Thompson JD, Koehl P, Ripp R, Poch O: **BAliBASE 3.0: latest developments of the multiple sequence alignment benchmark**. *Proteins* 2005, **61**:127-136.

40. de Bakker PI, Bateman A, Burke DF, Miguel RN, Mizuguchi K, Shi J, Shirai H, Blundell TL: **HOMSTRAD: adding sequence information to structure-based alignments of homologous protein families**. *Bioinformatics* 2001, **17**:748-749.

41. Raghava GP, Searle SM, Audley PC, Barber JD, Barton GJ: **OXBench: a benchmark for evaluation of protein multiple sequence alignment accuracy**. *BMC Bioinform* 2003, **4**:47.

42. Zemla A, Venclovas C, Moult J, Fidelis K: **Processing and analysis of CASP3 protein structure predictions**. *Proteins* **(Suppl. 3)**:1999:22-29.

43. Zhang Y, Skolnick J: **Scoring function for automated assessment of protein structure template quality**. *Proteins* 2004, **57**:702-710.

44. Ginalski K, Elofsson A, Fischer D, Rychlewski L: **3D-Jury: a simple approach to improve protein structure predictions**. *Bioinformatics* 2003, **19**:1015-1018.

45. Holm L, Sander C: **Dictionary of recurrent domains in protein structures**. *Proteins* 1998, **33**:88-96.

46. Armougom F, Moretti S, Keduas V, Notredame C: **The iRMSD: a local measure of sequence alignment accuracy using structural information**. *Bioinformatics* 2006, **22**:e35-e39.

47. Ginalski K, Grishin NV, Godzik A, Rychlewski L: **Practical lessons from protein structure prediction**. *Nucleic Acids Res* 2005, **33**:1874-1891.

48. Wallace IM, O'Sullivan O, Higgins DG, Notredame C: **M-Coffee: combining multiple sequence alignment methods with T-Coffee**. *Nucleic Acids Res* 2006, **34**:1692-1699.
• A meta-method that combines results of various alignment methods.

49. Katoh K, Toh H: **PartTree: an algorithm to build an approximate tree from a large number of unaligned sequences**. *Bioinformatics* 2007, **23**:372-374.

50. Pei J, Kim BH, Tang M, Grishin NV: **PROMALS web server for accurate multiple protein sequence alignments**. *Nucleic Acids Res* 2007, **35**:W649-W652.